

AI Clinical Workbench

An Intelligent Clinical Workflow Assistant for Reducing Healthcare Professionals' Documentation and Information Burden

Working product name: Clinical Workbench AI

Project type: Healthcare AI / Clinical Informatics / Human–AI Workflow

Primary objective: Build a practical clinical software layer that reduces documentation, information-organization and administrative work surrounding patient encounters while keeping healthcare professionals in control.

1. Project Overview

Clinical Workbench AI is an AI-assisted workspace designed to help healthcare professionals complete the work surrounding a patient encounter. It supports the clinician before, during and after the consultation by organizing patient context, capturing the encounter, drafting documentation, identifying downstream actions and preparing related clinical documents for review and approval.

The central product principle is: **One clinical encounter should not require the clinician to manually recreate the same information multiple times.**

2. The Problem

Clinical work involves much more than the direct conversation with a patient. A clinician may need to review previous records, write a consultation note, document findings, prepare referrals, give patient instructions, arrange follow-up and communicate information to other members of the care team.

Current ambient clinical documentation systems already demonstrate that AI can transcribe encounters and generate draft notes. This project therefore should not claim that AI medical scribes are new. Instead, it prototypes a broader encounter-to-workflow experience in which generated information becomes multiple useful, reviewable work products.

3. Current Technology Landscape

Modern healthcare AI already includes ambient documentation, clinical note generation, structured information extraction, patient-context preparation, medical coding and workflow-oriented clinical intelligence. Amazon Connect Health, for example, documents a workflow that captures and transcribes patient-clinician conversations and generates structured clinical documentation for professional review. [1]

AWS also provides a public sample combining patient context, ambient documentation, medical coding, review/approval and FHIR-based storage, with a local demonstration mode using synthetic Synthea data. [2]

The project should therefore be positioned as an engineering/research prototype exploring workflow integration and usability—not as a claim that the underlying concept is unprecedented.

4. Proposed Clinical Workflow

Stage	System support
1. Patient context	Summarize relevant history, recent encounters, investigations, medications, changes, outstanding investigations, referrals and follow-up.
2. Consultation	Capture or receive the clinician-patient conversation.
3. AI processing	Transcribe, structure and interpret the encounter within defined workflow boundaries.
4. Documentation	Generate a draft clinical note in the selected format.
5. Action extraction	Identify potential downstream actions such as repeat testing, referral, follow-up or patient instructions.
6. Document generation	Prepare drafts such as referral letters, patient instructions, handover summaries or discharge summaries.

Stage	System support
7. Review	Clinician checks, edits, accepts or rejects every AI-generated output.
8. Approval	Only approved information enters the intended clinical record/workflow.

5. Core Capabilities

- 1. Intelligent Patient Context** — Prepare a concise, source-grounded view of relevant patient information before the encounter.
- 2. Ambient Clinical Documentation** — Record or accept an encounter, transcribe the conversation and generate a structured clinical note.
- 3. AI Action Extraction** — Identify potential follow-up work stated during the encounter, such as repeat tests, referrals, follow-up appointments or patient instructions.
- 4. Automatic Clinical Document Generation** — Draft referral letters, patient instructions, handover summaries and discharge summaries.
- 5. Clinical Record Intelligence** — Let clinicians ask natural-language questions about the available patient record and return answers grounded in stored information.

6. Human-in-the-Loop Safety Model

The AI should generate and organize information; the healthcare professional remains responsible for reviewing and approving clinical outputs. AWS explicitly notes that ambient documentation outputs are probabilistic and must be reviewed by a trained medical professional before use in patient care. [1]

The prototype will avoid autonomous diagnosis, autonomous prescribing, irreversible orders, silent record modification and other actions that could create clinical consequences without professional review.

7. MVP Scope

1. Synthetic patient database
2. Patient profile and encounter history
3. Patient-context summary
4. Consultation audio recording/upload
5. Medical speech transcription
6. Clinical-note generation
7. Action extraction
8. Referral-letter draft
9. Patient-instruction draft
10. Follow-up task generation
11. Review, edit and approve interface
12. Encounter history
13. Audit trail

8. Technical Architecture

Clinician UI → AI Clinical Workbench → Patient Context / Consultation / Record Search → AI Orchestration → Transcription & Documentation / Information Retrieval → Workflow Engine → Draft Notes, Actions & Documents → Clinician Review → Approval → Structured Record

A provider-abstraction layer should keep AI components replaceable. The prototype can use free/open-source or locally available models where practical, while remaining compatible with external APIs later.

9. Cost Constraint

The project is designed around approximately **\$0 development spend**, excluding optional hosting. Development should use synthetic/public data, free or open-source software, local models where feasible and free service tiers where available. Real patient data should not be required.

The AWS unified clinical workflow sample demonstrates a local prototype mode using synthetic Synthea data without an AWS account. [2]

10. Evaluation Plan

Measure	Evaluation
Documentation time	Time required to produce the final note with and without the workbench.
Manual steps	Number of separate actions needed to complete the encounter workflow.
AI correction rate	How much generated content requires editing.
Note completeness	Whether relevant encounter information appears in the draft.
Action extraction accuracy	Whether requested follow-up actions are correctly identified.
Usability/workload	Perceived effort, usability and workflow friction.
Approval behavior	What users accept, edit or reject.

This evaluation is intended as a prototype/usability study, not evidence of clinical safety, diagnostic accuracy or patient-outcome improvement.

11. Research Question

Can an AI workbench reduce the number of manual steps required to complete the work surrounding a clinical encounter while keeping the clinician in control?

Secondary questions: Which workflow steps produce the greatest reduction in manual effort? How much editing is required before outputs are acceptable? Which downstream documents/actions can be reliably derived from one encounter? How does the interface affect perceived workload and usability?

12. Risks and Mitigation

Hallucination: source-grounded generation and clinician approval.

Missing information: distinguish information present, absent and inferred.

Incorrect action extraction: show supporting evidence and require approval before task creation.

Privacy: synthetic data during development.

Automation bias: make review and editing central to the workflow.

13. What This Project Is Not

It is not an AI doctor. It is not intended to diagnose patients or independently make treatment decisions. It is not simply a hospital management system. It is not limited to transcription. It is an AI-assisted workflow layer intended to reduce the work surrounding clinical encounters.

14. Long-Term Expansion

Future directions may include specialty-specific workflows, inpatient and emergency settings, nursing workflows, FHIR/SMART-on-FHIR integration, EHR integration, clinical knowledge retrieval, structured data extraction and broader care-team collaboration.

15. Final Project Vision

“An AI workbench that helps healthcare professionals get the work around a patient encounter done.”

The immediate goal is not to replace clinical judgment. The goal is to remove repetitive information and documentation work so healthcare professionals can spend more time on the patient and less time recreating information across the workflow.

16. Selected References

- [1]** Amazon Connect Health. Ambient documentation and getting-started documentation, 2026.
- [2]** AWS Samples. Unified Clinical Workflow on Amazon Connect Health; MIT-0 reference implementation and synthetic-data demo mode.
- [3]** Guo, Y. Human-Centered Evaluation of Ambient Artificial Intelligence Clinical Documentation, University of California, Irvine, 2026.
- [4]** Wieben et al. Protocol for an EHR-embedded pragmatic randomized controlled trial of Ambient AI to Reduce Nursing Staff Documentation Time, 2026.